Team 70

Phone Addiction Predictor

Execution Lead

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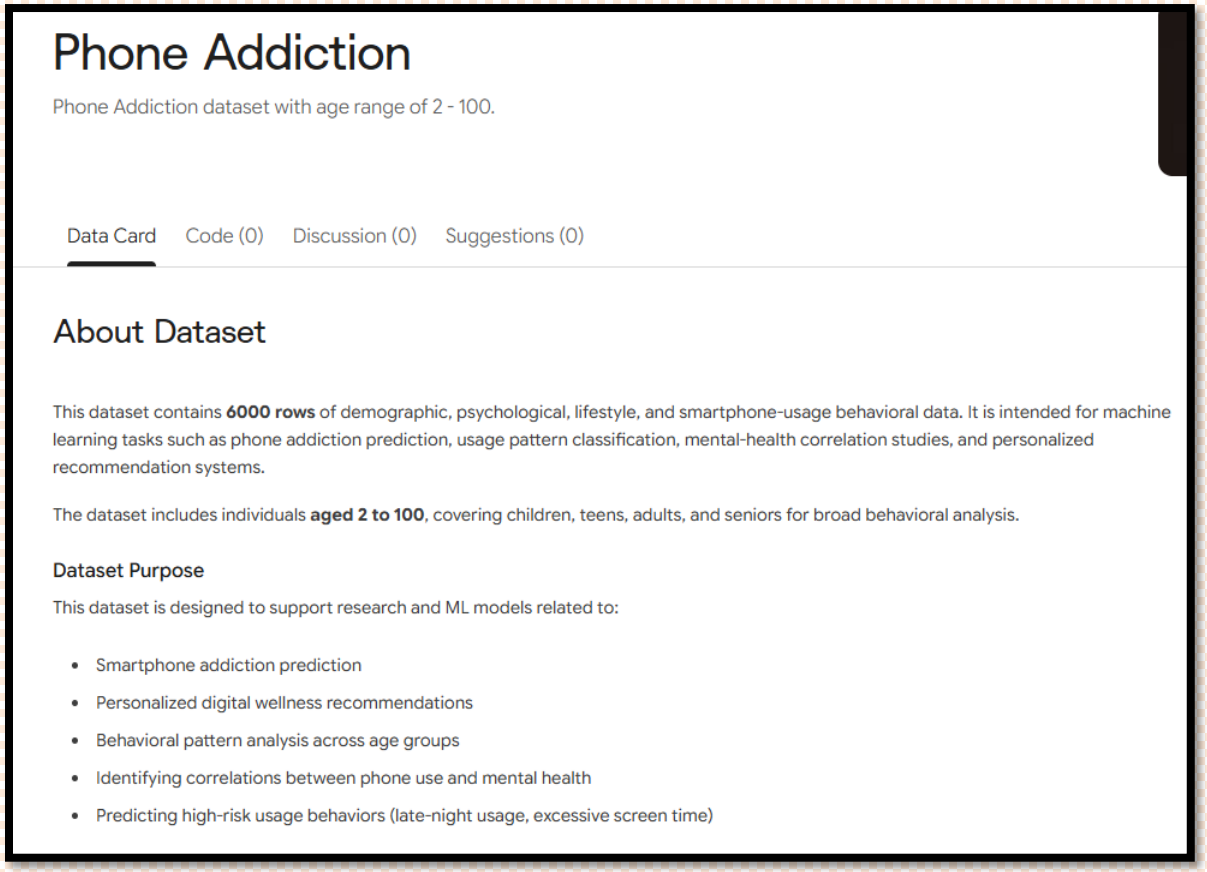
github.com/priyam-here/phone-addiction-predictor

Problem & Data

WHAT & WHY

We aim to predict the phone addiction level using demographic, psychological, and smartphone-usage behavioral data such as hours of usage, purpose of usage, social interaction, etc.

> **Challenge:** The dataset contains severe categorical noise (2,726 unique locations) and an artificial mathematical ceiling capping addiction levels at 10.



Phone Addiction

Phone Addiction dataset with age range of 2 - 100.

Data Card | Code (0) | Discussion (0) | Suggestions (0)

About Dataset

This dataset contains **6000 rows** of demographic, psychological, lifestyle, and smartphone-usage behavioral data. It is intended for machine learning tasks such as phone addiction prediction, usage pattern classification, mental-health correlation studies, and personalized recommendation systems.

The dataset includes individuals **aged 2 to 100**, covering children, teens, adults, and seniors for broad behavioral analysis.

Dataset Purpose

This dataset is designed to support research and ML models related to:

- Smartphone addiction prediction
- Personalized digital wellness recommendations
- Behavioral pattern analysis across age groups
- Identifying correlations between phone use and mental health
- Predicting high-risk usage behaviors (late-night usage, excessive screen time)

DATASET PROFILE

- Size: 6,000 rows
- Structure: 22 feature columns

Dataset Link:

https://www.kaggle.com/datasets/kirtansukhadiya/phone-addiction/data

Approach & Pipeline

LIBRARIES USED

pandas, scikit-learn, matplotlib, seaborn, os



1. Raw Data

Load 6,000 survey rows



2. Clean

Drop unique locations
And remove the artificial cap of Depression Level 10



3. Encode Data

StandardScaler & Dummies to convert categorical data into numerical columns



4. Train

Linear Regression Model and evaluate on various metrics

Results & Demo

MODEL 1: CLEANED DATA

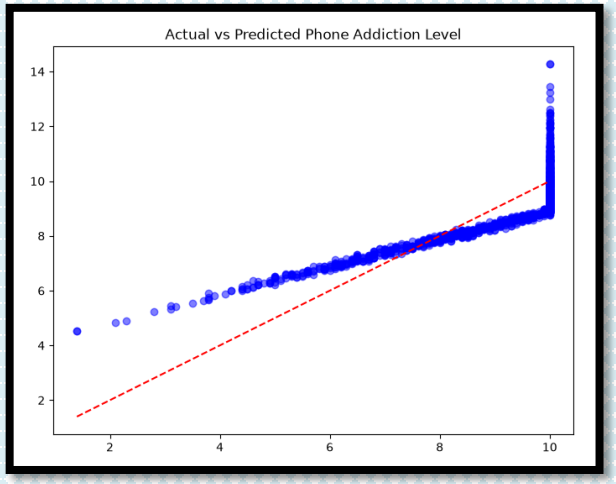
R^2: 0.7325

Location removed.
Artificial cap of Addiction = 10 remains.

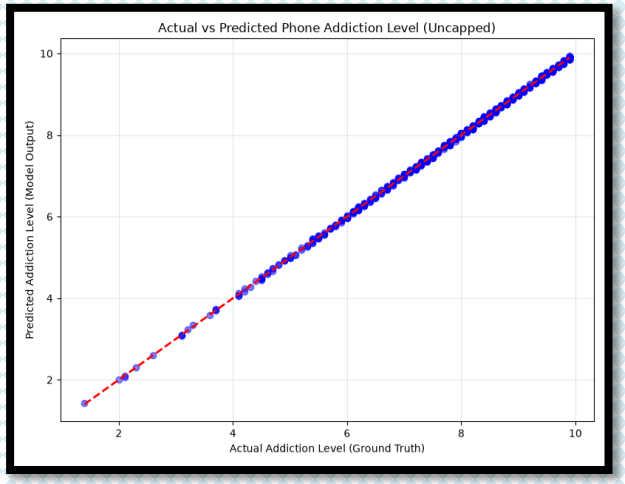
MODEL 2: UNCAPPED

R^2: 0.9997

Final: True underlying formula revealed.



Actual vs Predicted Phone Addiction Level Before Removing the Artificial Cap of Addiction <=10



Actual vs Predicted Phone Addiction Level After Removing the Artificial Cap Of Addiction <= 10

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Training Linear Regression model...

--- MODEL RESULTS (TRAIN vs TEST) ---
R-squared (R2)  | Train: 0.9997 | Test: 0.9997
Mean Sq. Error  | Train: 0.0008 | Test: 0.0009
Mean Abs. Error | Train: 0.0241 | Test: 0.0252
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Conclusion: Train and Test metrics are nearly identical,
proving NO OVERFITTING.

Generating visualizations in 'outputs/' directory...
Pipeline complete.
```

Learnings & AI Use

WHAT WORKED & DIDN'T

Failed: Catastrophic Overfitting on 'Location': Initially, we included the Location column. Because it contained 2,726 unique cities, one-hot encoding it exploded the dataset's dimensionality. The model memorized the training data but catastrophically failed on the test data (yielding an R^2score of -8.4×10^24).

The Artificial Data Ceiling: Our fixed baseline model achieved 73% accuracy, but we discovered the predictions were bottlenecked by an artificial cap where all maximum addiction scores were clamped strictly at 10.

Worked: Identifying the **artificial metric cap** at 10. Removing these 3,000+ rows allowed the model to achieve 99.9% accuracy; effectively meaning that the model was able to **reverse engineer the formula** used for the Addiction Level from the dataset itself.

Dimensionality Reduction: Dropping Location and Name columns to avoid overfitting.

Scaling using StandardScaler made the coefficients comparable.

WHO DID WHAT

Aniruddh: Data Prep & Scaling, Execution Lead

Poojitha: Training & Metrics, Video Lead

Dhruvil: Plots & Visuals of the output, PPT Lead

Priyam: Main Pipeline code & Documentation, Github repository work

AI DISCLOSURE

Gemini: Diagnosed the dimensionality error caused by the Location column, generated parts of the code of data_prep.py and plots.py, and structured the README. All code was reviewed and tested locally.

Generated the base version of the PPT which was then heavily modified by us according to our needs and standards.